

Collision Resolution Techniques (Open Addressing & Chaining) — Data Structures

Definition: A **Collision** occurs in hashing when a hash function maps two distinct keys k_1 and k_2 to the exact same table index ($h(k_1) = h(k_2)$). Collision resolution techniques resolve this conflict.

1. Detailed Technical Explanation

COLLISION RESOLUTION STRATEGIES

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| 1. OPEN ADDRESSING (Closed Hashing) | 2. SEPARATE CHAINING (Open Hashing) |
| - Linear Probing | - Array of Linked Lists |
| - Quadratic Probing | |
| - Double Hashing | |

2. Technique 1: Open Addressing

All keys are stored directly inside the hash table array. If slot $h(k)$ is occupied, systematic probing finds the next vacant slot.

1. Linear Probing

Probes slots sequentially with an offset of 1:

$$h(k, i) = (h'(k) + i) \bmod m \quad \text{for } i = 0, 1, 2, \dots, m-1$$

- Drawback:** Suffers from **Primary Clustering** (long contiguous blocks of occupied slots build up, degrading search time).

2. Quadratic Probing

Probes slots using a quadratic polynomial:

$$h(k, i) = (h'(k) + c_1 * i + c_2 * i^2) \bmod m$$

- Advantage:** Eliminates primary clustering; but can cause **Secondary Clustering** (keys with same initial hash follow identical probe sequences).

3. Double Hashing (Best Open Addressing Technique)

Uses two independent hash functions $h_1(k)$ and $h_2(k)$:

$$h(k, i) = (h_1(k) + i * h_2(k)) \bmod m$$

- Rule: $h_2(k)$ must never evaluate to 0 and must be relatively prime to m .
- Example: $h_1(k) = k \bmod 11$ and $h_2(k) = 7 - (k \bmod 7)$.

3. Technique 2: Separate Chaining

Each slot in the hash table points to the head of a **Linked List** storing all colliding keys mapped to that index.

```
Hash Table Array
Index 0: NULL
Index 1: [ 12 ] -> [ 23 ] -> [ 34 ] -> NULL (Collisions at index 1 chained)
Index 2: [ 13 ] -> NULL
Index 3: NULL
```

Separate Chaining vs Open Addressing Comparison:

Feature	Separate Chaining	Open Addressing
Storage Structure	Array of Linked Lists.	Single Array (Keys stored in slots).
Table Capacity	Can store $> m$ elements ($\alpha > 1$).	Limited to table size m ($\alpha \leq 1$).
Deletion	Simple node removal from linked list.	Complex (requires DELETED dummy markers).
Cache Performance	Poor cache locality due to linked list pointers.	Excellent cache locality (contiguous array).

4. Quick Recall Flow

```
Collision (h(k1) == h(k2)) -> Open Addressing: Linear (i), Quadratic (i^2), Double Hashing (i*h2) |
Separate Chaining: Linked List buckets
```